

Data Engineer Contribution — Stage 3

Retention Radar — Customer Churn Prediction (ATS)

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For: Team Final Report and Predictive Modelling folder — Stage 3 submission

1. Files for the Predictive Modelling folder

These two files should go in the folder alongside the model architecture document and the trained model itself:

- Stage3_DataEngineer_QA_Validation_Report.docx — a full report covering seven separate checks on the data hand-off
- stage3_data_consistency_audit.py — the script used to run those checks, so anyone can re-run it if the data changes

Here is a plain summary of what was checked and what was found:

Check	Result
Row counts & split	6,741 full → 5,392 train / 1,349 test — matches spec. PASS
Churn stratification	26.57% full / 26.58% train / 26.54% test. PASS
Train/test leakage	0 overlapping rows. PASS
Scaler fit-on-train-only	Independently refit — matches shipped files exactly. PASS
Missing values / duplicates	0 across all 5 hand-off files. PASS
Feature-set consistency	All 3 feature sets used in Stage 3 notebook map cleanly to Stage 2 columns and significance testing. PASS

In short: all seven checks passed. The data is clean, and the model was trained on a reliable, well-prepared dataset.

2. Key Findings — a note on data quality

The model was trained and tested using data that had already been carefully checked. There were 6,741 customer records in total, split into a training set and a testing set (80% and 20%), with the split done in a way that kept the churn rate balanced across both. There was no overlap between the two sets, and the scaling of numeric values was done correctly, using only the training data. This matters because it means the model's reported accuracy and ROC-AUC score of 0.817 can be trusted as a fair reflection of how it would perform on new, unseen customers, not something inflated by a flawed setup.

3. Limitations and Proposed Solutions

Limitation 1 — In this dataset, contract type and internet service type did not turn out to be strong predictors of churn, even though they usually are in telecom data. This is worth calling

out clearly so the team does not mistake it for an error in the model — it is simply a pattern found in this dataset.

What we suggest: treat contract type as a secondary factor rather than a main driver when recommending retention strategies and instead lean on the factors that were consistently shown to matter — tenure, monthly charges, whether the customer has dependents, and whether they are a senior citizen.

Limitation 2 — Two features, contract length in months and total number of services, were kept in the data mainly to support customer segmentation work, even though on their own they showed very little connection to churn. If these show up with some importance in the model's results, it should be read as a segmentation signal rather than a strong churn driver.

What we suggest: when presenting feature importance results, make this distinction clear so readers do not overinterpret these two features.

Limitation 3 — The model was tested using a single 80/20 split of the data rather than testing it multiple times across different splits (a method called cross-validation). This means the results are solid, but there is a small chance they could shift slightly if tested a different way.

What we suggest: this is a reasonable limitation given the time available for this internship project. For future work, running the model across five different splits (5-fold cross-validation) would confirm the accuracy and ROC-AUC figures hold up consistently.